

Danmarks
Tekniske
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02417 - TIME SERIES ANALYSIS

Assignment 3

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Contents

1	Problem 1 Stationarity	2
2	Problem 2 Predicting monthly solar power	8
3	Problem 3 An ARX model for the heating of a box	12
4	Use of AI	19
4.1	Declaration of Generative AI and AI-assisted technologies in the writing process .	19

1 Problem 1 Stationarity

1.1

We have the $AR(2)$ process

$$X_t + \phi_1 X_{t-1} + \phi_2 X_{t-2} = \epsilon_t \quad (1.1)$$

where ϵ_t is a white noise process with $\sigma_\epsilon = 1$. In Figure 1, we have plotted five realizations of the process with $n = 200$ observations using the coefficients $\phi_1 = -0.6$ and $\phi_2 = 0.5$.

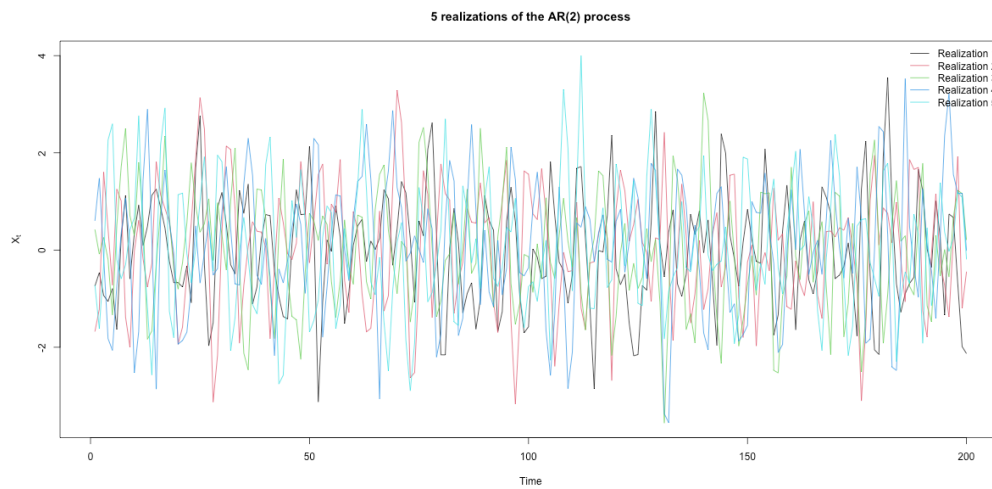


Figure 1: Five simulated realizations of the process 1.1 with $n = 200$, $\phi_1 = -0.6$, and $\phi_2 = 0.5$.

1.2

We calculate and plot the empirical ACF's for the five simulations in section 1 alongside the theoretical ACF for process 1.1 in R, using the `acf` and `ARMAacf` functions respectively.

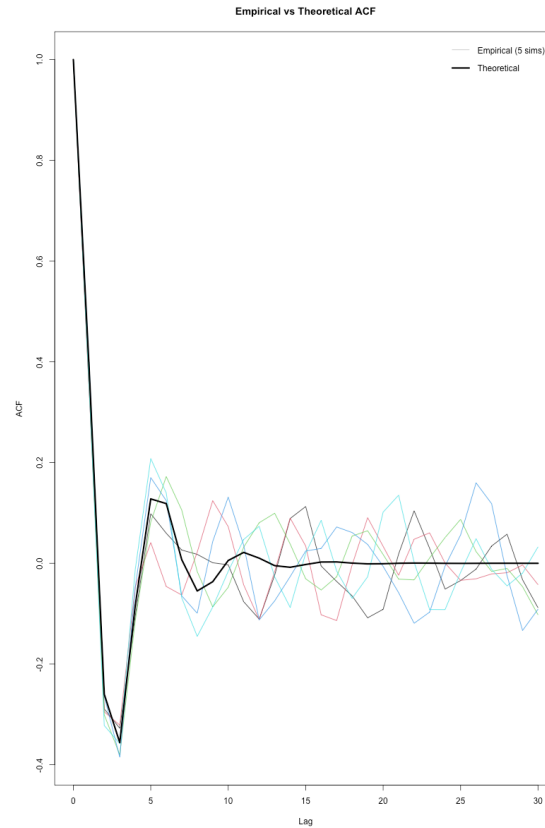


Figure 2: Empirical vs theoretical autocorrelation functions for simulations of process 1.1. Seed 123 in R.

The theoretical ACF shows a damped sinusoidal decay. This behavior is consistent with a stationary AR(2) process according to the Golden Table of ARMA identification. The ACF approaches zero around lag 15, indicating stationarity. This can be verified using Theorem 5.9. The characteristic polynomial is

$$\phi(z) = 1 - 0.6z + 0.5z^2 = 0. \quad (1.2)$$

Solving for the roots gives

$$z = 0.6 \pm i\sqrt{1.64}. \quad (1.3)$$

We compute the inverse roots

$$\frac{1}{z} = \frac{0.6 \pm i\sqrt{1.64}}{0.6^2 + 1.64} = \frac{0.6 \pm i\sqrt{1.64}}{2}. \quad (1.4)$$

Their magnitude is

$$\left| \frac{1}{z} \right| = \frac{1}{\sqrt{2}} < 1. \quad (1.5)$$

Thus, the roots of $\phi(z^{-1}) = 0$ lie within the unit circle, and by Theorem 5.9 the process is stationary. In other words, the theoretical ACF plot matches the expected pattern. We observe that the ACFs of the simulations follow the theoretical ACF relatively closely throughout the entire range. In particular, for the first lags, up to approximately lag 5, they all follow the theoretical ACF almost identically. After this, they begin to oscillate around the theoretical ACF with small phase differences. These deviations in the empirical ACFs are most likely due to noise introduced by the finite number of observations, $n = 200$.

1.3

In this simulation we use the coefficients $\phi_1 = -0.6$ and $\phi_2 = -0.3$. We analyze the characteristics of this process as before by finding the roots of the characteristic polynomial:

$$\phi(z) = 1 - 0.6z - 0.3z^2 = 0 \quad (1.6)$$

$$z = \frac{0.6 \pm \sqrt{0.36 + 1.2}}{-0.6} = \frac{0.6 \pm \sqrt{1.56}}{-0.6} \quad (1.7)$$

$$z_1 \approx -3.08, \quad z_2 \approx 1.08 \quad (1.8)$$

$$\frac{1}{z_1} \approx -0.325, \quad \frac{1}{z_2} \approx 0.926 \quad (1.9)$$

Thus, we obtain two real roots, and the inverse roots lie inside the unit circle, and the process is stationary in accordance with theorem 5.9. This results in an ACF that exhibits exponential decay towards zero. From the simulation in R, this is exactly what we observe for both the theoretical and empirical ACFs, as shown in Figure 3. The simulations all follow the theoretical ACF and each other closely at early lags but start to deviate progressively at later lags, which is comparable to the previous plot 2. In broad terms, however, the five simulations in Figure 3 demonstrate very similar behavior. Furthermore, the time series plot displays no trend and oscillates around zero, exhibiting constant variance.

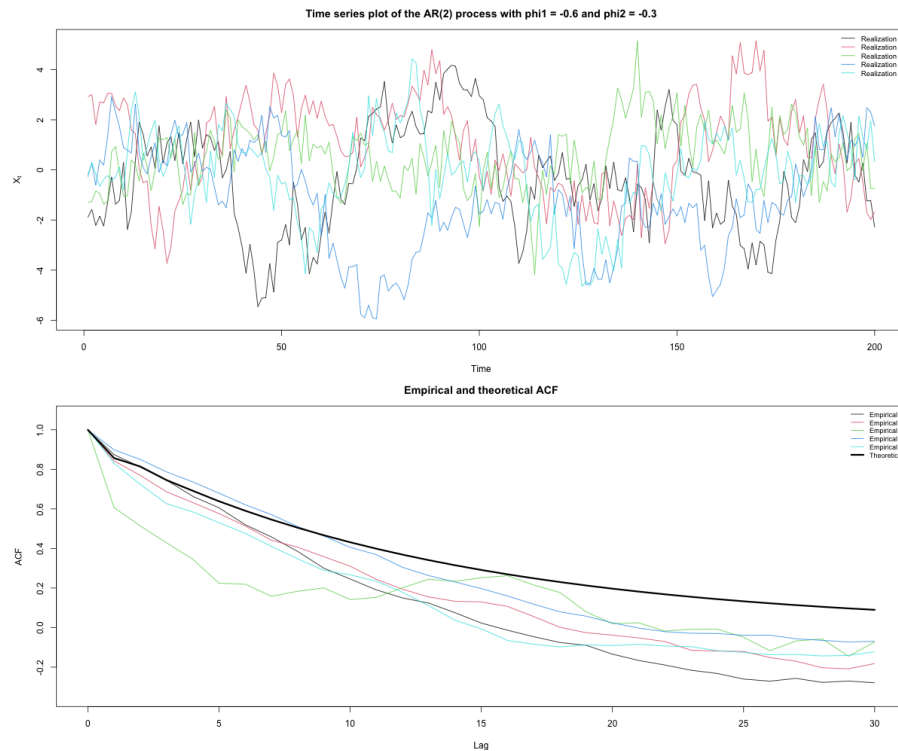


Figure 3: Time series, theoretical/empirical autocorrelation function for simulations of process 1.1 with $\phi_1 = -0.6$, and $\phi_2 = -0.3$.

1.4

In this simulation we use the coefficients $\phi_1 = 0.6$, and $\phi_2 = -0.3$. The roots of the characteristic polynomial are:

$$\phi(z) = 1 + 0.6z - 0.3z^2 = 0 \quad (1.10)$$

$$z = \frac{-0.6 \pm \sqrt{0.36 + 1.2}}{-0.6} = \frac{-0.6 \pm \sqrt{1.56}}{-0.6} \quad (1.11)$$

$$z_1 \approx 3.08, \quad z_2 \approx -1.08 \quad (1.12)$$

$$\frac{1}{z_1} \approx 0.325, \quad \frac{1}{z_2} \approx -0.926 \quad (1.13)$$

The inverse roots are inside the unit circle, hence the process is stationary, which is apparent from the decay observed in the ACF plot in Figure 4. Furthermore, the process exhibits oscillatory behavior. Across the simulations, we see once again that for smaller lags, the empirical ACFs follow the theoretical ACF with similar amplitudes and identical phases. From lag 5 onwards, however, we observe variations in the empirical ACFs, most notably in the amplitudes of the oscillations. Around lag 20, some of the simulations even appear to increase in amplitude, which conflicts with our analysis of the roots of the characteristic polynomial, which showed that the process is stationary. The time series, however, confirms that the process is stationary, as the realizations oscillate around zero with a seemingly stable variance.

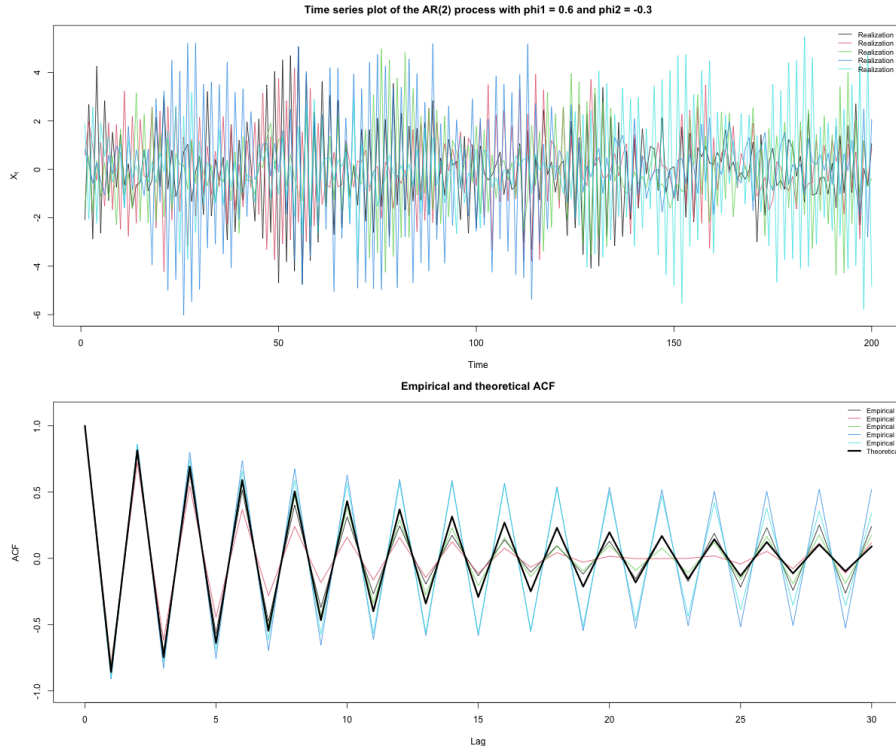


Figure 4: Time series, theoretical/empirical autocorrelation function for simulations of process 1.1 with $\phi_1 = 0.6$, and $\phi_2 = -0.3$.

1.5

Using coefficients $\phi_1 = -0.7$, and $\phi_2 = -0.3$, we obtain the following roots:

$$\phi(z) = 1 - 0.7z - 0.3z^2 = 0 \quad (1.14)$$

$$z = \frac{0.7 \pm \sqrt{0.49 + 1.2}}{-0.6} = \frac{0.7 \pm \sqrt{1.69}}{-0.6} \quad (1.15)$$

$$z_1 \approx -3.33, \quad z_2 = 1 \quad (1.16)$$

$$\frac{1}{z_1} \approx -0.3, \quad \frac{1}{z_2} = 1 \quad (1.17)$$

As one of the roots is exactly equal to 1 and therefore lies on the unit circle, the process is not stationary according to theorem 5.9. In Figure 5, we see that the realizations in the time series plot increasingly drift away from zero, some in the positive and others in the negative direction. This also indicates that the process is not stationary, as the variance increases over time. In the ACF plot, we clearly observe that the theoretical ACF does not decay towards zero, which further confirms that the process is not stationary. The empirical ACFs do not follow the theoretical ACF and decay at different rates, which can give a misleading impression of the process characteristics.

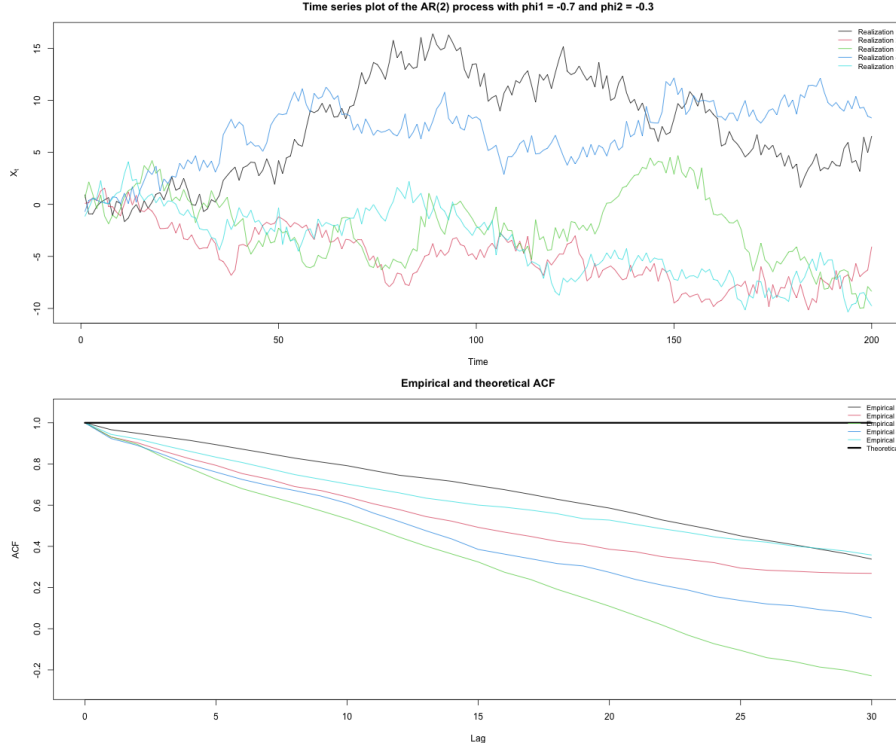


Figure 5: Time series, theoretical/empirical autocorrelation function for process 1.1 with $\phi_1 = -0.70$, and $\phi_2 = -0.3$.

1.6

In this simulation we use the coefficients $\phi_1 = -0.75$, and $\phi_2 = -0.3$.

$$\phi(z) = 1 - 0.75z - 0.3z^2 = 0 \quad (1.18)$$

$$z = \frac{0.75 \pm \sqrt{1.76}}{-0.6} \quad (1.19)$$

$$z_1 \approx -3.46, \quad z_2 = 0.96 \quad (1.20)$$

$$\frac{1}{z_1} \approx -0.29, \quad \frac{1}{z_2} \approx 1.04 \quad (1.21)$$

Since one of the inverse roots lies outside the unit circle, the process is non-stationary according to Theorem 5.9. This is evident in the time series plot in Figure 6, where the realizations explode after $t = 100$, both in the negative and positive directions, clearly indicating a non-stationary process. The theoretical ACF also reflects non-stationarity, as it grows and does not decay towards zero. The empirical ACFs, however, decay towards zero, but since they do not follow the theoretical ACF, they give a misleading impression of the stationarity of the process.

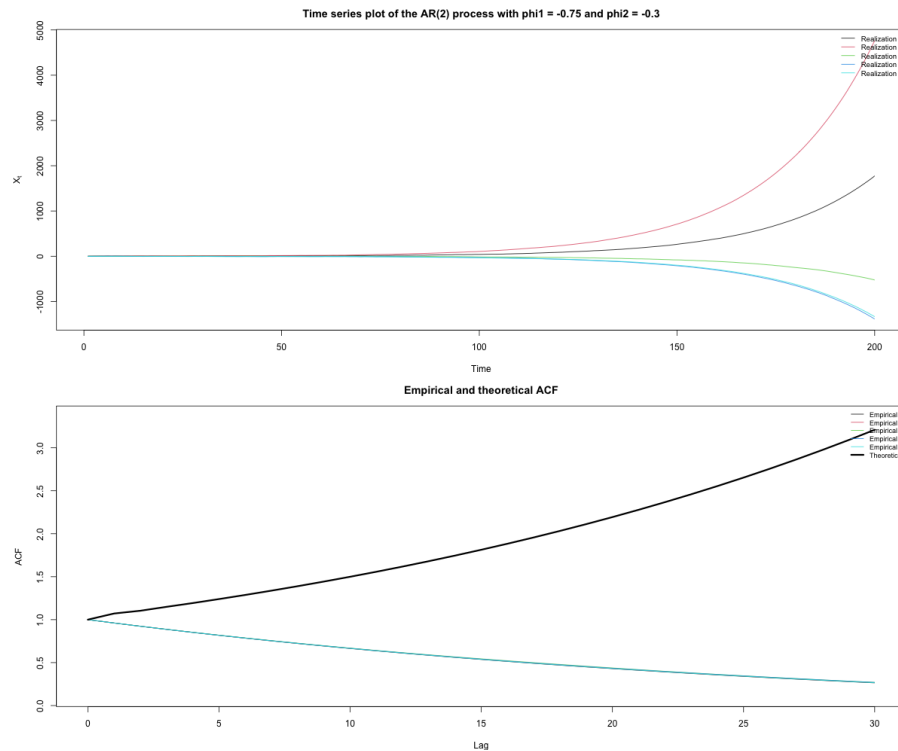


Figure 6: Time series, theoretical/empirical autocorrelation function for process 1.1 with $\phi_1 = -0.75$, and $\phi_2 = -0.3$.

1.7

The ACF plots provide a good overview of the autocorrelation between lags and can give insight into the behavior of the process, such as oscillations and decay, which reflect the underlying dynamics and can be used, to some extent, to assess stationarity. However, we also observe that the empirical ACFs can give a misleading impression of the stationarity of the process. In Question 1.4, the empirical ACFs increase again after a certain number of lags, even though our analysis showed that the process is stationary according to Theorem 5.9. Similarly, the empirical ACFs in 1.5 and 1.6 show decay towards zero and do not follow the theoretical ACF. In contrast, plotting the time series provides direct insight into the process dynamics. It can be used to evaluate whether the process exhibits trends or stable or unstable variance, and thus provides complementary insight into stationarity. Therefore, to obtain a complete understanding of the process, including its stationarity, we recommend always plotting the time series data and not relying solely on the ACF.

2 Problem 2 Predicting monthly solar power

The monthly energy production from the solar PV plant is modeled by

$$(1 + \phi_1 B)(1 + \Phi_1 B^{12})(\log(Y_t) - \mu) = \varepsilon_t, \quad (2.1)$$

where the parameters are given as $\phi_1 = -0.38$, $\Phi_1 = -0.94$, $\mu = 5.72$, and the noise variance is $\sigma_\varepsilon^2 = 0.22^2 = 0.0484$.

We define

$$X_t = \log(Y_t) - \mu, \quad (2.2)$$

so the model can be written as

$$(1 + \phi_1 B)(1 + \Phi_1 B^{12})X_t = \varepsilon_t. \quad (2.3)$$

Residuals and model validation

Expanding the model gives

$$X_t + \phi_1 X_{t-1} + \Phi_1 X_{t-12} + \phi_1 \Phi_1 X_{t-13} = \varepsilon_t. \quad (2.4)$$

This means that the one-step-ahead predictor is

$$\hat{X}_{t+1|t} = -\phi_1 X_t - \Phi_1 X_{t-11} - \phi_1 \Phi_1 X_{t-12}. \quad (2.5)$$

Inserting the numerical values gives

$$\hat{X}_{t+1|t} = 0.38X_t + 0.94X_{t-11} - 0.3572X_{t-12}. \quad (2.6)$$

The residuals are therefore calculated as

$$\hat{\varepsilon}_{t+1|t} = X_{t+1} - \hat{X}_{t+1|t}. \quad (2.7)$$

The residual diagnostics are shown in Figure 7. Overall, the residuals look acceptable. The residual series fluctuates around zero without any obvious trend, and the residual ACF does not show strong significant autocorrelation. A normal Q–Q plot also suggests that the residuals are reasonably close to normal, although there is a small deviation in the upper tail. This is not surprising, since the sample is quite short.

The residual diagnostics do not show strong remaining autocorrelation, and the residuals appear reasonably close to normal. Therefore, the i.i.d. assumption seems acceptable for this model, although the conclusion should be interpreted with some caution because the sample is relatively small.

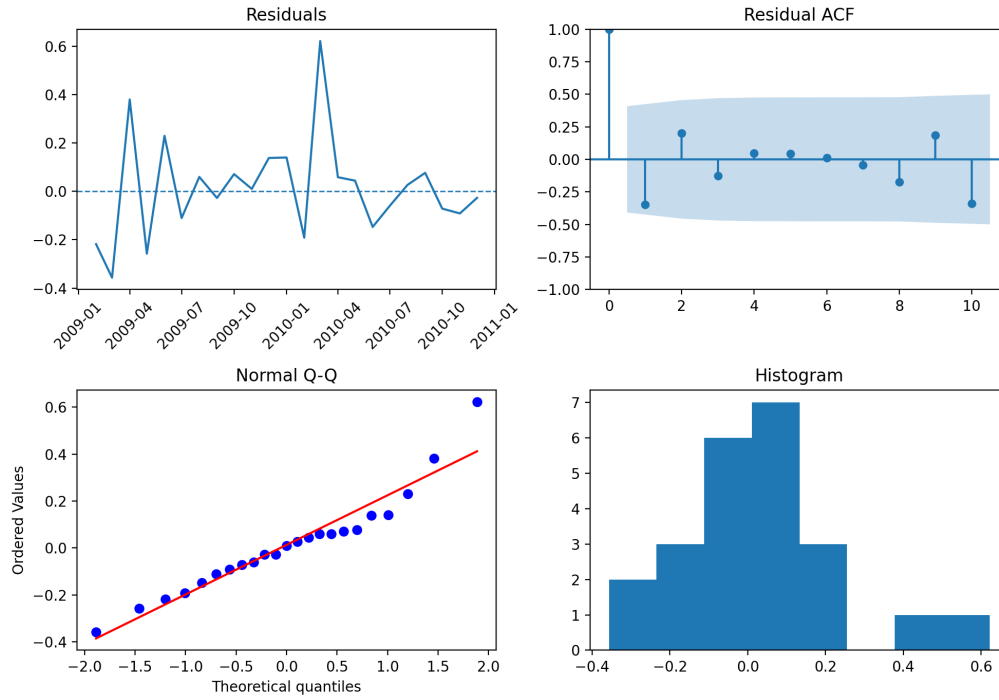


Figure 7: Residual diagnostics for the seasonal AR model.

Twelve months ahead prediction

The goal is to predict the next 12 months of energy production from time $t = 36$. The forecasts are first computed on the log-scale using the recursive predictor

$$\hat{X}_{36+k|36} = -\phi_1 \hat{X}_{36+k-1|36} - \Phi_1 X_{36+k-12} - \phi_1 \Phi_1 X_{36+k-13}, \quad k = 1, \dots, 12. \quad (2.8)$$

After this, the forecasts are transformed back to the original scale as

$$\hat{Y}_{36+k|36} = \exp(\mu + \hat{X}_{36+k|36}). \quad (2.9)$$

The forecasted values are shown in Table 1. The predicted pattern follows the same yearly seasonality as in the observed data. The production is low in the winter months, increases through spring, reaches its highest level in summer, and then decreases again towards the end of the year. This seems physically reasonable for a solar power plant.

Table 1: 12-month ahead forecasts and 95% prediction intervals for monthly solar production.

Month	Forecast [MWh]	Lower 95%	Upper 95%
2011-01	101.3	65.8	155.9
2011-02	147.2	92.8	233.5
2011-03	156.9	98.5	249.9
2011-04	142.6	89.4	227.2
2011-05	355.3	222.9	566.3
2011-06	538.0	337.6	857.5
2011-07	611.2	383.5	974.2
2011-08	631.9	396.5	1007.2
2011-09	630.1	395.3	1004.3
2011-10	508.2	318.8	809.9
2011-11	375.0	235.3	597.7
2011-12	280.5	176.0	447.1

Figure 8 shows the observed time series together with the 12-month forecast and prediction intervals. The forecast extends the seasonal pattern in a smooth way, which is what we would expect from this kind of seasonal AR model.

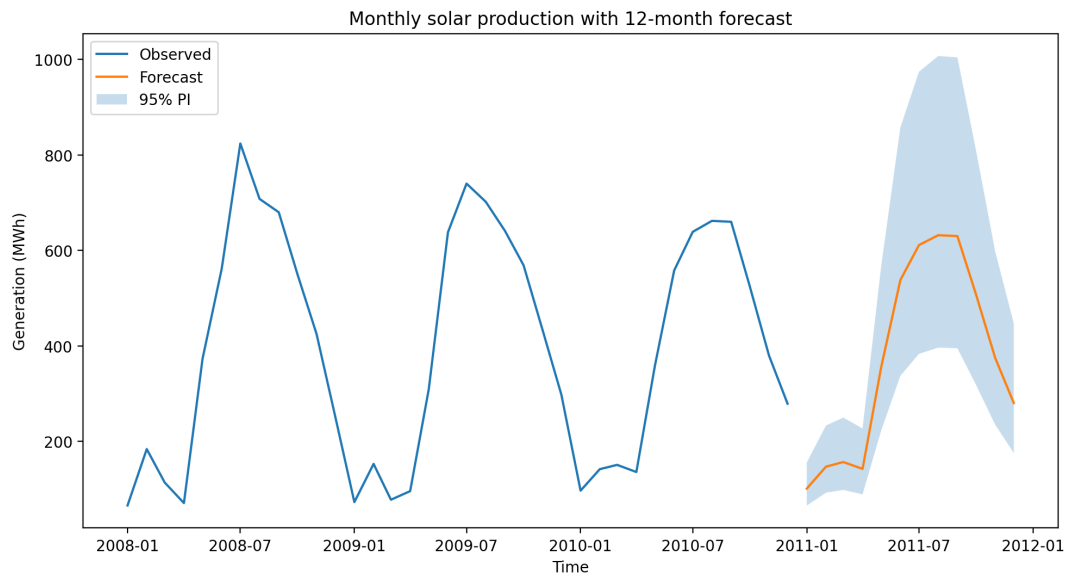


Figure 8: Observed monthly solar production together with 12-month ahead forecasts and 95% prediction intervals.

Prediction intervals

The 95% prediction intervals were calculated using only the AR(1) part, as stated in the assignment. Since the seasonal lagged terms are already observed for horizons up to 12 months, the forecast uncertainty is driven by the non-seasonal AR(1) term.

Because $\phi_1 = -0.38$, the AR coefficient used in the recursion is

$$a = -\phi_1 = 0.38. \quad (2.10)$$

The innovation variance given in the assignment is

$$\sigma_\varepsilon^2 = 0.22^2 = 0.0484. \quad (2.11)$$

The forecast error variance on the X_t scale is then

$$\text{Var}(X_{36+k} - \hat{X}_{36+k|36}) = \sigma_\varepsilon^2 \sum_{j=0}^{k-1} a^{2j} = 0.0484 \sum_{j=0}^{k-1} 0.38^{2j}. \quad (2.12)$$

This can also be written as

$$\text{Var}(X_{36+k} - \hat{X}_{36+k|36}) = 0.0484 \cdot \frac{1 - 0.38^{2k}}{1 - 0.38^2}. \quad (2.13)$$

The 95% prediction limits on the log-scale are therefore

$$\hat{X}_{36+k|36} \pm 1.96 \sqrt{\text{Var}(X_{36+k} - \hat{X}_{36+k|36})}, \quad (2.14)$$

and these are then transformed back to the original scale by exponentiation.

The intervals are wider in the summer months, where the expected production is highest. Since the model is built on the log-scale and then transformed back to MWh, the intervals become asymmetric on the original scale, with a longer upper side.

Discussion

In general, the forecast looks reasonable and captures the main seasonal behavior of the solar plant well. The model predicts low production in winter and high production in summer, which matches the structure of the observed data. So in that sense, the forecast can be trusted as a description of the overall annual pattern.

However, I would be more careful about trusting the exact monthly values. The dataset is short, with only 36 observations, and the model only uses past production values. It does not include explanatory variables such as weather conditions, cloud cover, irradiation, or possible operational changes in the plant. Because of that, the model is not able to react to unusual future conditions.

The prediction intervals seem reasonable as a first approximation, but they should still be interpreted with some caution. The sample is short, and parameter uncertainty is not included, so the true uncertainty may be a bit larger than what is shown here. Also, because the model is estimated on the log-scale, the intervals become asymmetric after transforming back to MWh.

Overall, the model seems useful for getting a rough 12-month forecast and for describing the expected seasonal pattern, but it should not be seen as a very precise forecasting tool for individual months.

3 Problem 3 An ARX model for the heating of a box

3.1

From the plot in figure 9, we see the three time series, we see the P_h fluctuating between approximately 0W to 100W. We see the temperature difference T_{delta} between the inside and outside of the fluctuating in approximately 12 – 26°C, and always positive, indicating that the inside is always warmer than the outside of the box.

Lastly we see the solar radiation G_v , showing obvious daytime pulses, with nighttime lows, which is very consistent with winter conditions in Belgium.

A very clear relationship can be observed between P_h and G_v , where the heating power rises during the low periods of the solar radiation, and rises during high time in solar radiation. This make intuitive sense, since solar radiation creates heat, that is then not necessary for the heater to create.



Figure 9: Plot of the time series of P_h, T_{delta}, G_v , thick vertical lines represent the hour 00:00, and bottom plot x-axis labels correspond to both above plots.

3.3

The plots in figure 10 show a scatter plot of P_h with T_{delta} and P_h with G_v and a CCF plot of P_h with T_{delta} and P_h with G_v .

In the P_h with T_{delta} plots, we can see that there seems to be a positive correlation, a larger temperature difference, means that more heating is required. This is evident in both the scatter and the CCF plot, with the scatter trending up and to the right, and the CCF showing large correlations on small lags, and decaying off thereafter.

In the P_h with G_v plots, we can see a strong negative correlation, more solar radiation leads to reduced heating power. Evident in both the scatter plot, and the CCF with large negatives at lags 0-2, and decaying off.

From these plots however, it is not necessarily seen directly, whether the dynamics of this system can be captures using a linear model.

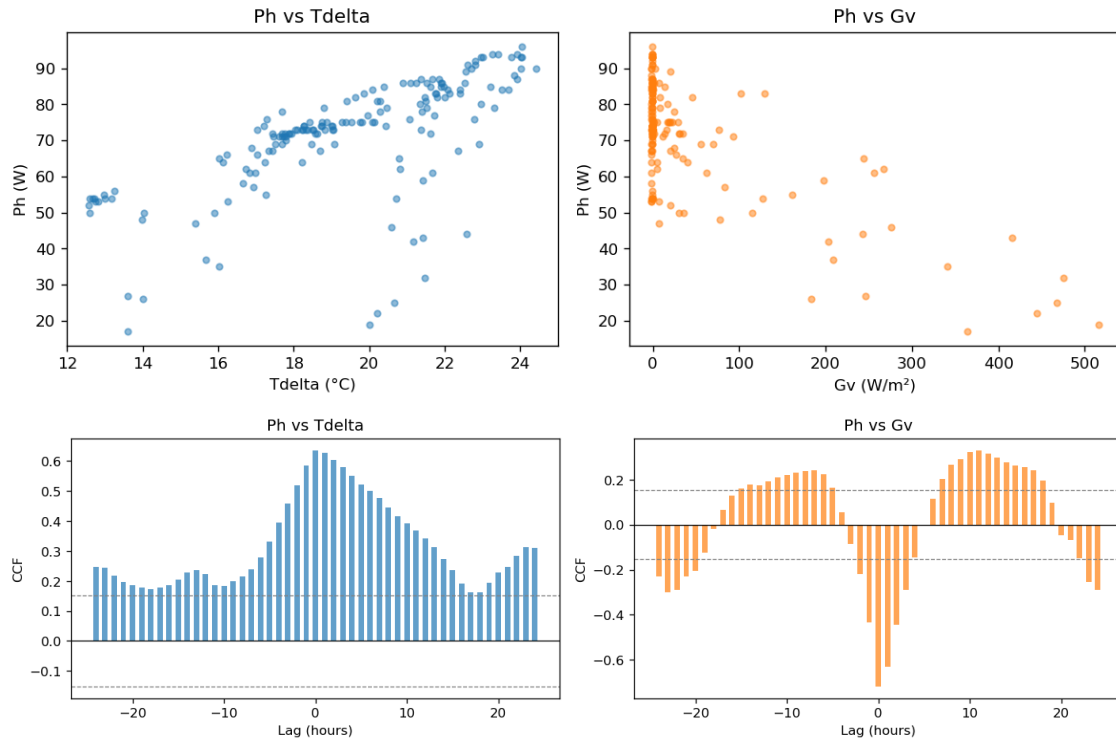


Figure 10: Plot of P_h/T_{delta} and P_h/G_v in both scatter plot(top) and CCF(bottom)

3.4

Figure 11 shows the impulse response from T_{delta} and G_v to P_h . The impulse response shown on the left, the $T_{delta} \rightarrow P_h$, shows an oscillating coefficient in the first 4 lags, followed by some fairly low positive coefficients, and finally 2 even lower, negative coefficients. The behavior seems fairly random, and we therefore decide the impulse response does not give concrete enough evidence of a relationship.

The impulse response shown on the right, the $G_v \rightarrow P_h$, shows a large negative response at lag 0, and decays with following lags. Pointing towards solar radiation reduces the heating power. Again, this is intuitive since the solar radiation will heat the box walls.

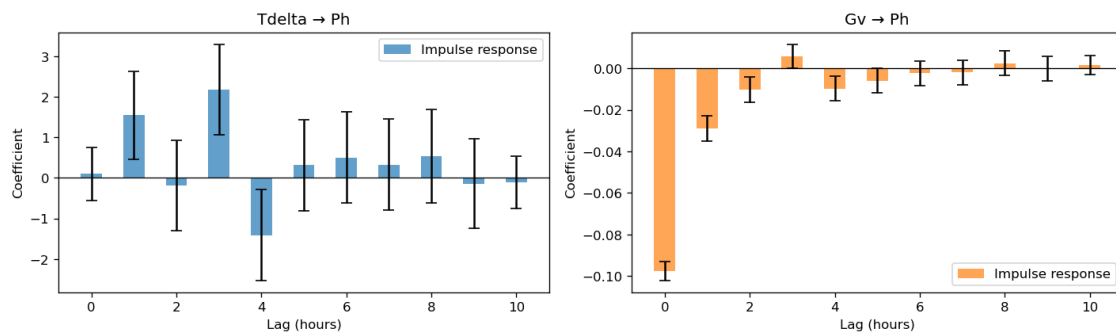


Figure 11: Plot of impuls response in both $T_{delta} \rightarrow P_h$ and $G_v \rightarrow P_h$

3.5

We fit the linear regression model

$$P_{h,t} = \omega_1 T_{delta,t} + \omega_2 G_{v,t} + \epsilon_t \quad (3.1)$$

Analyzing the plots in figure 12, we see the CCF of residuals vs the inputs, the residuals vs T_{delta} show a clear correlation at negative lags, thus meaning that past residuals correlate with future T_{delta} . Suggesting a transfer function with dynamics could be included. The CCF of residuals with G_v also show large correlations at 0 and negative lags, with an oscillating pattern, hinting of a day/night-time cycle, that this model fails to capture.

Further, we can see that in the one-step predictions, that the regression follows the general pattern, but still lags behind in some of the sharper peaks or drops. This suggests that a better fitting model could be created.

Moving further to the residuals over time, we see that these are clearly not just random noise. This is a strong sign of autocorrelation in the errors.

The ACF of the residuals confirms this, with significant autocorrelations at lags 1, 2, 3, and 4, that land outside of the 95% confidence band. Thus meaning that knowing a residual, will tell the model something about the next one, again this motivates a better fitting model.

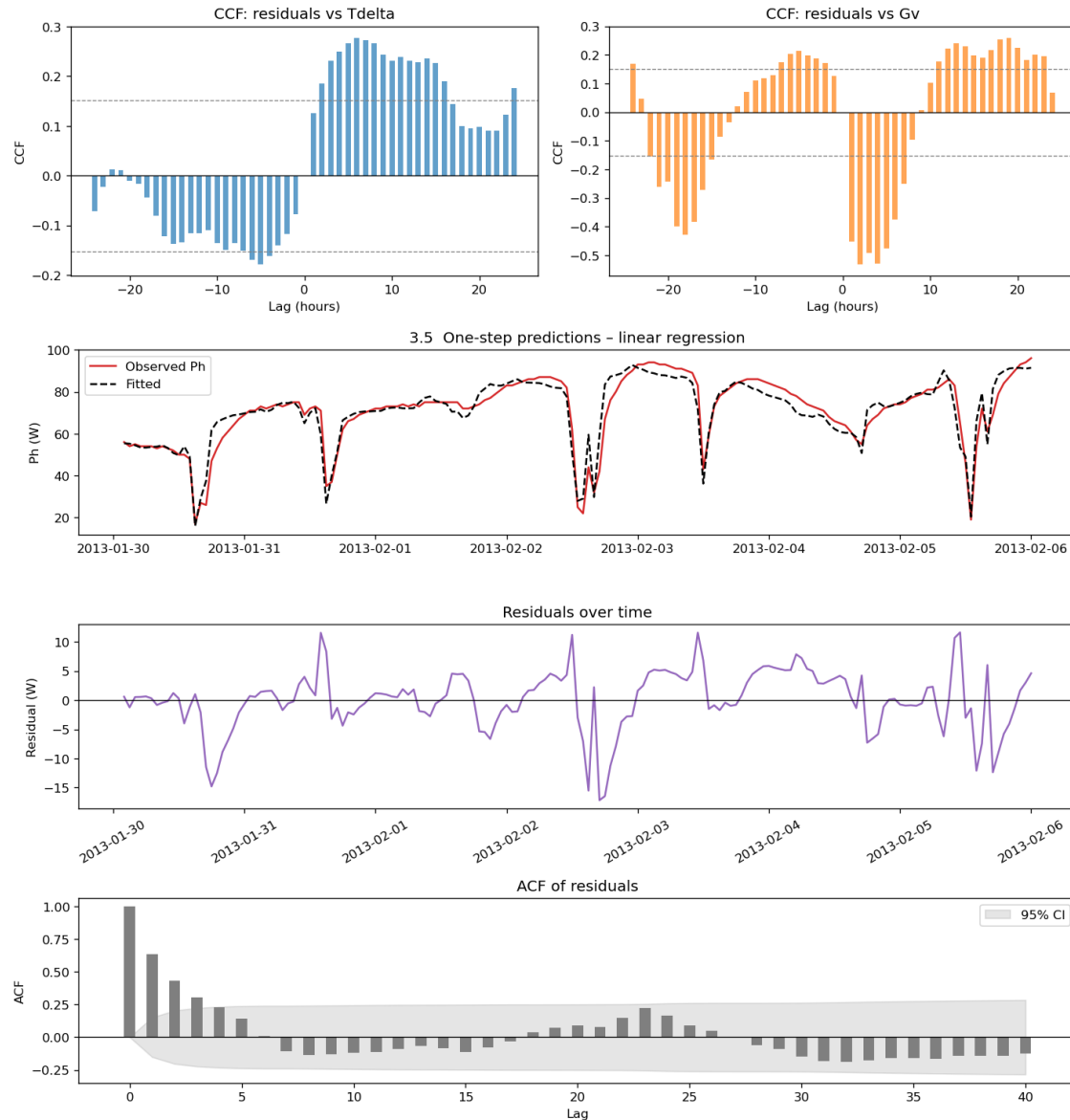


Figure 12: Plot of CCF with residuals and T_{delta} and G_v (top) and linear regression with time series and ACF.

3.6

We now fit the first-order ARX model

$$P_{h,t} = \phi_1 P_{h,t-1} + \omega_1 T_{delta,t} + \omega_2 G_{v,t} + \epsilon_t \quad (3.2)$$

Looking at the plots in figure 13 we can see a clear improvement from adding the first order ARX. We now see the ACF of the residuals drop quickly at lag 1, into the 95% confidence band. And the observed and predicted regression seem to be fitting better together, specifically sharp peaks and drops are a closer fit.

The residuals are also looking significantly more like noise, notice the amplitude band has dropped from peaks at around 10 and -15 to about 7 and -10. However there is still some peaks and drops in the most extreme cases.

However there is still room for improvement in the model, through increasing the model order.

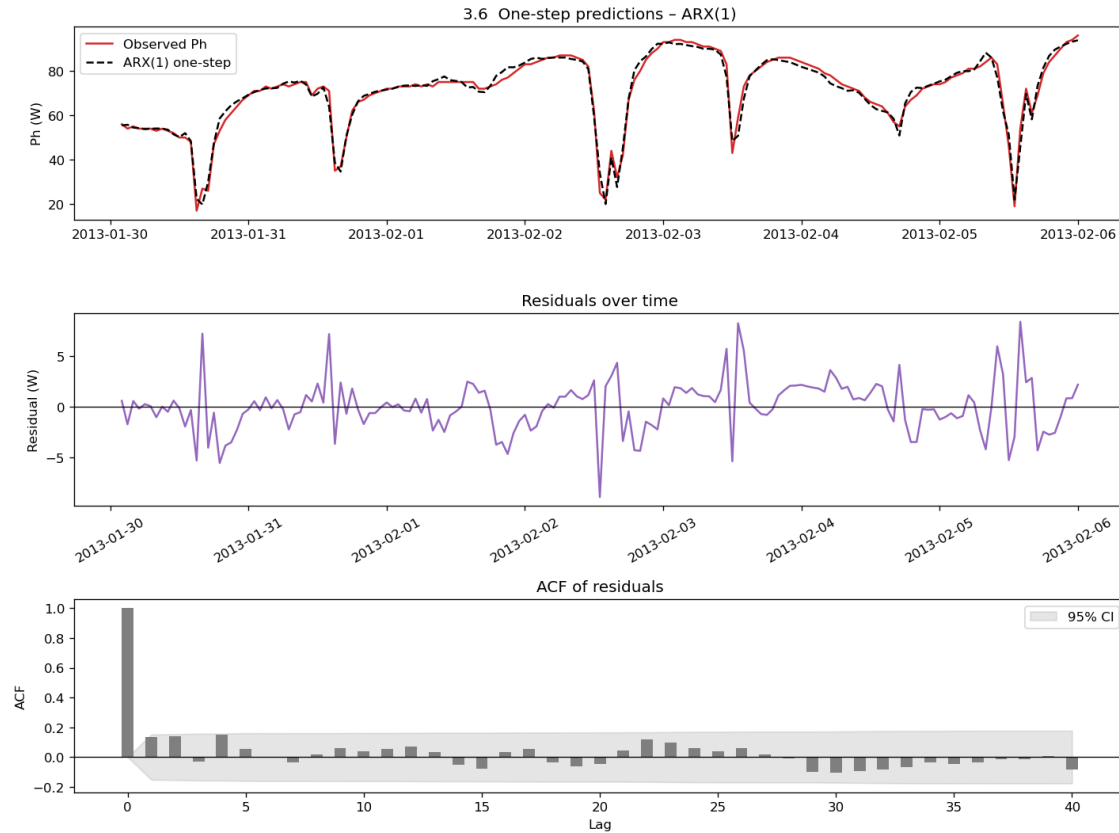


Figure 13: Plot of residuals ACF with ARX

3.7

We now increase the model order.

$$P_{h,t} = \phi_1 P_{h,t1} \phi_2 P_{h,t2} + \omega_{1,0} T_{delta,t} + \omega_{1,1} T_{delta,t1} + \omega_{2,0} G_{v,t} + \omega_{2,1} G_{v,t1} + \epsilon_t \quad (3.3)$$

The plot of BIC and AIC vs the model order in figure 14 show both AIC and BIC having a minimum at order 6. Therefore this is a good candidate for a chosen model. However we do see that the BIC levels off around model order 3, and remains fairly flat, until it starts to rise again with the model complexity. Therefore an argument could be made for choosing model 3. And that the AIC remains flat after 6, therefore there are no real gain (or loss) from choosing later orders.

We see that the two differ in the ways the penalize the model, BIC penalizes more heavily on complexity. We can see the gap between the two increases as the model complexity rises.

Everything taken into account, the large drop from 0 to 1, and the diminishing returns after 4, i we would argue that model 3 or 4 would be justified, having a good fit without overfitting. But 6 is the formal optimal choice.

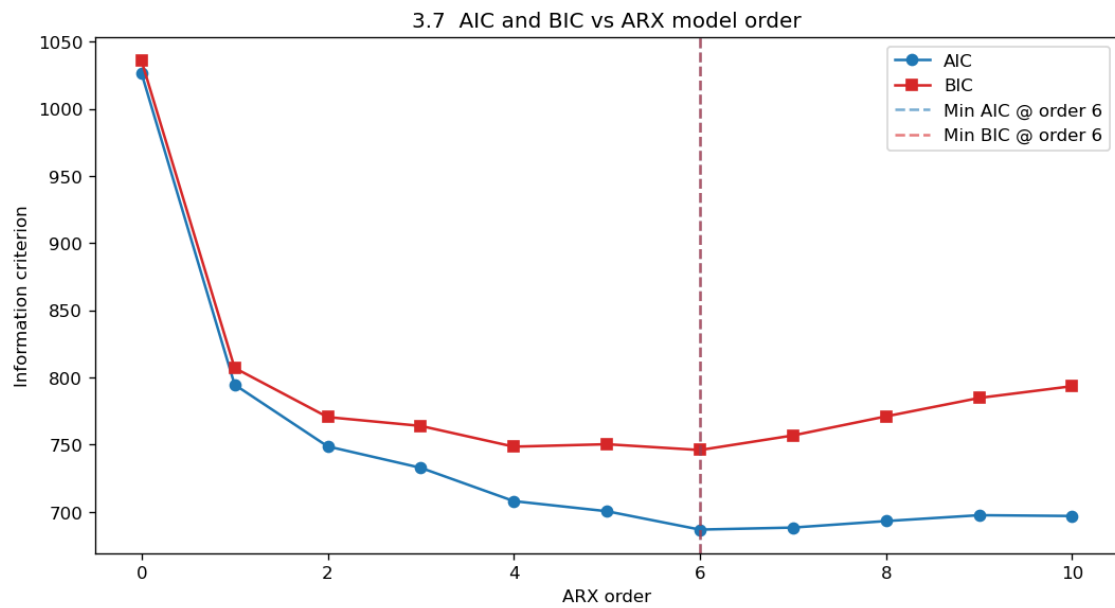


Figure 14: Plot of AIC and BIC on the ARX model

3.8

Figure 15 shows the RMSE, and we see a large drop from 0 to 1, and 1 to 2, and then a clear minimum at 3. After order 3, the RMSE rises slightly, and flattens from order 6 to 10.

This is not in agreeance with the BIC and AIC, suggesting that the orders 4-6 may be capturing some noise in the training set, which is a sign of overfitting.

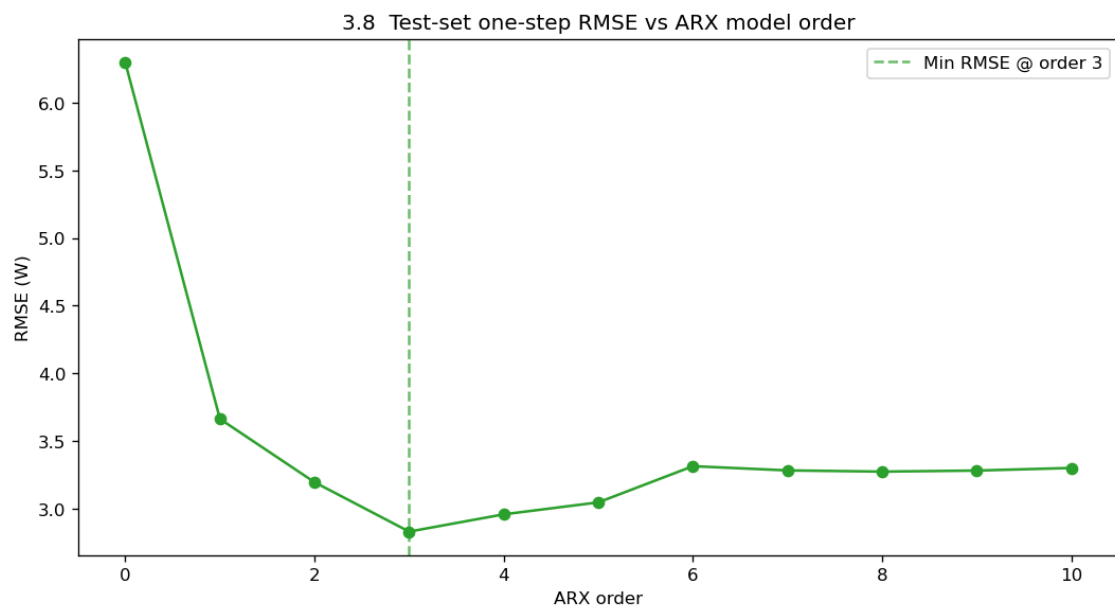


Figure 15: Plot of model order with one-step predictions with RMSE.

3.9

We see that the observed P_h in figure 16 is tracked very well by the simulation, especially the dips and peaks are tracked better with the final model. The error plot further shows a well performing model. Worth noting is the spike near the training/test data split, one reason for this may be the model needing to "adjust" to the test data, and this together with a large dip in the observed data, may have introduced a more erroneous result.

One big reason the multi-step approach works for this case is, that the inputs of T_{delta} and G_v are directly fed, and do not compound in uncertainty. Therefore the simulation is only really worried about the AR dynamics.

In an operational setting, the prediction could be feasible, provided with knowledge of the input variables T_{delta} and G_v . T_{delta} could in practice be known from a weather forecast, and G_v from previous measurements of solar radiation in the area at that time.

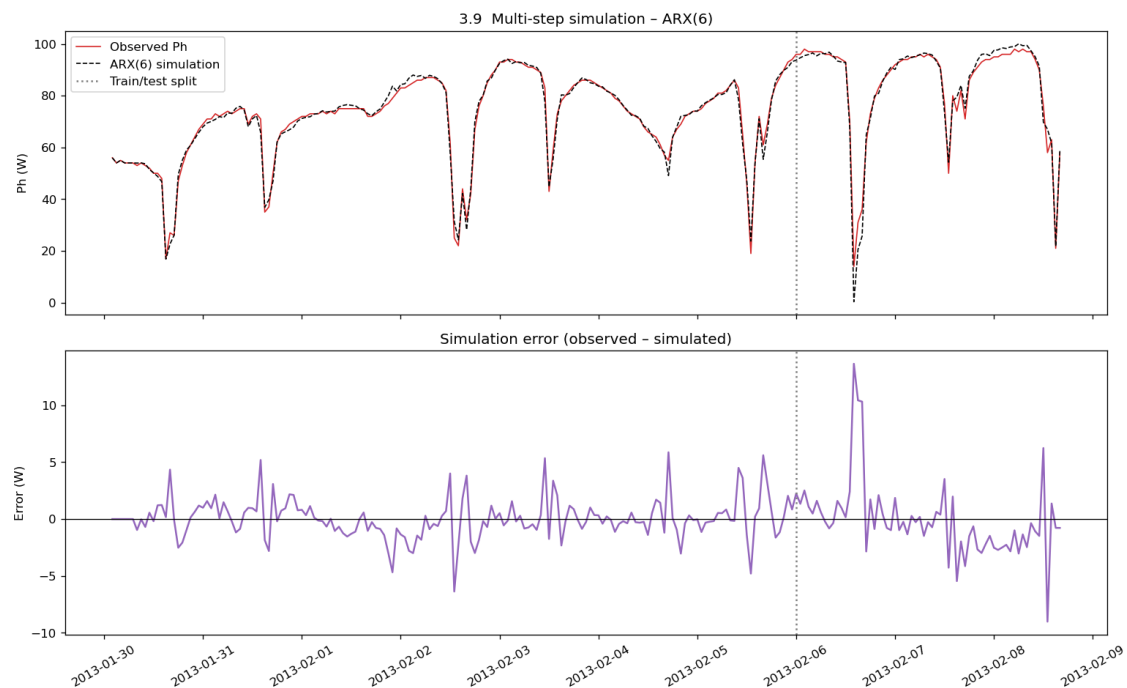


Figure 16: Plot of multi-step simulation, using model order 6

3.10

First we looked at the three time series, they showed clear day- and nighttime patterns. P_h is positively correlated with T_{delta} , and negatively correlated with G_v . Then we tried fitting a linear regression, which already captured the main effects, but leave a lot of autocorrelation in the residuals. We then added AR lags, which significantly reduced the residual autocorrelation. We then tested AIC and BIC on multiple model orders, and found nr. 6 was the best. Then we found the most optimal model order according to one-step RMSE, which found it was order 3. Then we ran a multi-step simulation, which showed a stable model. Overall an ARX model provides a practical model for the heating prediction in this building.

4 Use of AI

4.1 Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Github Copilot and chatGPT in order to

- develop/debug code,
- improve grammar,
- make corrections,
- assist with LaTeX formatting.

After using this tool/service, the authors reviewed and edited the content as needed and take(s) full responsibility for the content of the publication